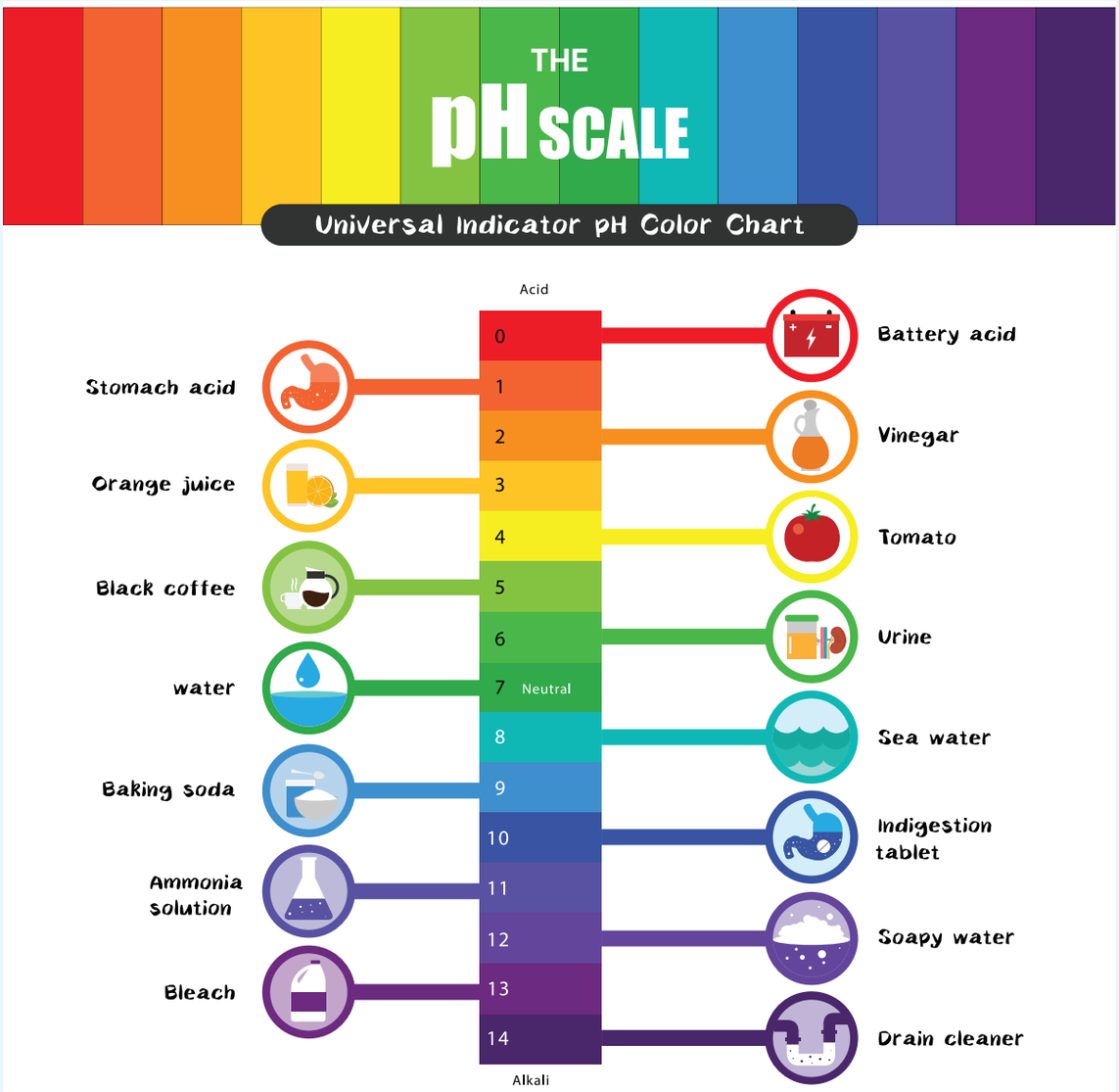


Figure 8.1 The pH scale

$$pH + pOH = -\log_{10}[H_3O^+] - \log_{10}[OH^-]$$

$$= -\left(\log_{10}[H_3O^+] + \log_{10}[OH^-]\right)$$

$$pH+pOH = -\log_{10}[H_3O^+][OH^-] \qquad [\because \log a + \log b = \log ab]$$

We know that $[H_3O^+][OH^-] = K_W$

$$\Rightarrow pH+pOH = -\log_{10} K_W$$

$$\Rightarrow pH+pOH = pK_W \qquad [\because pK_W = -\log_{10} K_w] \qquad .....(8.8)$$

at 25°C, the ionic product of water, $K_W = 1\times10^{-14}$

$$pK_W = -\log_{10} 10^{-14} = 14 \log_{10} 10$$

$$= 14$$

$\therefore$ (8.7) $\Rightarrow$ $\therefore$ At 25°C, pH + pOH = 14

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